LARGE ALMOST-EQUIDISTANT SETS FROM TRIANGLE-FREE STRONGLY REGULAR GRAPHS

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ABSTRACT. A set of points in $\mathbb{R}^d$ is almost-equidistant if among any three of them some two are at distance 1. Let $f(d)$ be the largest size of an almost-equidistant set in $\mathbb{R}^d$. The best lower bound previously recorded for every $d \geq 10$ was $f(d) \geq 2d+4$, and the only structured ingredient in the literature was the 16-point set of Larman and Rogers in $\mathbb{R}^5$ built from the Clebsch graph. We observe that every regular triangle-free graph on n vertices whose second largest eigenvalue $\lambda_2 > 0$ has multiplicity m yields an almost-equidistant set of n points on a sphere in $\mathbb{R}^{n-1-m}$, and that two spherical almost-equidistant sets can be combined by an orthogonal direct sum whenever their squared radii sum to 1. Applied to the Hoffman–Singleton, Gewirtz, M_{22} and Higman–Sims graphs, and complemented by an extension lemma based on Delsarte-tight cocliques, this gives

$$f(10) \geq 26, \qquad f(20) \geq 58, \qquad f(21) \geq 78, \qquad f(22) \geq 100,$$
$$f(22+j) \geq 100 + 2j \quad (1 \leq j \leq 8), \qquad f(31) \geq 120,$$

hence $f(d) > 2d + 4$ for all $20 \leq d \leq 57$, and $f(22) \geq 100 > 4.5 \cdot 22$. The Higman–Sims set is a two-distance almost-equidistant set of 100 points in $\mathbb{R}^{22}$, answering a question of Polyanskii for $n = 100$. All constructions are certified by an exact, dependency-free verification script.

1. INTRODUCTION

For a positive integer d, a finite set $V \subset \mathbb{R}^d$ is *almost-equidistant* if among any three points of V some two are at Euclidean distance 1. Let $f(d)$ denote the largest cardinality of an almost-equidistant set in $\mathbb{R}^d$. The problem was raised by Z. Lángi and is a Euclidean analogue of Erdős's question on almost-orthogonal vectors, solved by Rosenfeld [14]. The vertex set of the Moser spindle shows $f(2) \geq 7$; Bezdek, Naszódi and Visy [3] proved $f(2) = 7$ and Györey proved $f(3) = 10$ (see [1]).

Balko, Pór, Scheucher, Swanepoel and Valtr [1] made a systematic study of f. They proved $f(d) \leq O(d^{3/2})$, later improved to $O(d^{13/9})$ by Polyanskii [12] and to $O(d^{4/3})$ by Kupavskii, Mustafa and Swanepoel [10] (see also [13]). On the lower bound side, [1] gives $f(d) \geq 2d + 4$ for all $d \geq 3$ by an explicit construction, and, using the 16-point set of Larman and Rogers [11] in $\mathbb{R}^5$ derived from the Clebsch graph, $f(5) \geq 16$, $f(6) \geq 18$, $f(7) \geq 20$ and $f(9) \geq f(8) \geq 24$. Table 1 of [1] records $2d + 4$ as the best known lower bound for every $d \geq 10$, and the conjecture $f(d) = O(d)$ is stated in [12, Conjecture 12]. Polyanskii [13, Problems 18 and 19] asked for the largest almost-equidistant set that is at the same time a two-distance set, and reduced this to a question about triangle-free graphs whose second largest eigenvalue has large multiplicity.

The purpose of this note is to point out that the Larman–Rogers construction is the first member of a family, and to record the consequences. The Clebsch graph is one of the seven known triangle-free strongly regular graphs (the pentagon, the Petersen, Clebsch, Hoffman–Singleton, Gewirtz, M_{22} and Higman–Sims graphs), and the argument of Larman and Rogers applies verbatim to each of them. More generally, it applies to every regular triangle-free graph through the multiplicity of its second largest eigenvalue.

1.1. **Theorem.** *Let H be a connected k-regular triangle-free graph on n vertices whose second largest adjacency eigenvalue λ_2 is positive and has multiplicity m. Then there is an almost-equidistant set of n points in $\mathbb{R}^{n-1-m}$, all lying on a sphere of radius r with*

$$r^2 = \frac{1}{2} + \frac{k-\lambda_2}{2n\lambda_2}.$$

Only two distances occur in this set, namely 1 and $\sqrt{1+1/\lambda_2}$.

Combined with an orthogonal direct-sum construction (Lemma 2.1) and an extension lemma (Lemma 4.1), Theorem 1.1 gives the following.

1.2. **Theorem.** *The following lower bounds hold:*

d	10	20	21	22	23	24	25	26	27	28	29	30	31
$f(d) \geq$	26	58	78	100	102	104	106	108	110	112	114	116	120
previous	24	44	46	48	50	52	54	56	58	60	62	64	66

Consequently, since f is non-decreasing, $f(d) \geq 120 > 2d+4$ for $31 \leq d \leq 57$.

The bound $f(22) \geq 100$ comes from the Higman–Sims graph alone; $f(31) \geq 120$ adds to it the 20 vertices of a regular unit 9-simplex and its antipodal image, placed in an orthogonal $\mathbb{R}^9$. The two-distance set of 100 points in $\mathbb{R}^{22}$ also shows that the minimum of $\operatorname{rank}(A-\lambda_2 I)$ over triangle-free graphs on 100 vertices with $\lambda_2 > 0$, the quantity in [13, Problem 19], is at most 23.

We stress that the ingredients are classical: the spectral embedding of a graph into an eigenspace, the Delsarte–Hoffman bound for cocliques, and the seven known triangle-free strongly regular graphs. The contribution is the observation that they combine to produce almost-equidistant sets, and the exact bookkeeping needed to turn this into verified numbers. Everything in Theorem 1.2 is certified by a standard-library Python script that rebuilds the graphs from the binary Golay code, checks all identities in integer and rational arithmetic, and tests every triple of points; the script is reproduced in Appendix A.

2. Spherical almost-equidistant sets and direct sums

Throughout, $\langle\cdot,\cdot\rangle$ is the standard inner product. For a set of points on a sphere of radius r centred at the origin, two points p, q satisfy $\|p-q\|^2 = 2r^2 - 2\langle p, q\rangle$, so

$$\|p-q\| = 1 \iff \langle p, q\rangle = r^2 - \tfrac{1}{2}. \tag{1}$$

It is convenient to normalise: for $c \in (-1,1)$ let $h(d,c)$ be the largest size of a set of unit vectors in $\mathbb{R}^d$ such that among any three of them two have inner product exactly c. Scaling by $r = (2(1-c))^{-1/2}$ shows that $h(d,c)$ is the largest size of an almost-equidistant set on the sphere of radius r in $\mathbb{R}^d$, where $c = 1 - 1/(2r^2)$. Rosenfeld [14] proved $h(d,0) = 2d$, and Bezdek and Lángi [2] proved $h(d,c) \leq 2d+2$ for all $c \leq 0$, with equality for $c = -1/d$.

2.1. **Lemma** (Direct sum)**.** *Let $P \subset \mathbb{R}^{d_1}$ and $Q \subset \mathbb{R}^{d_2}$ be almost-equidistant sets such that $\|p\| = r_1$ for all $p \in P$, $\|q\| = r_2$ for all $q \in Q$, and $r_1^2 + r_2^2 = 1$. Then $(P \times \{0\}) \cup (\{0\} \times Q)$ is an almost-equidistant set of $|P| + |Q|$ points in $\mathbb{R}^{d_1+d_2}$.*

Proof. Write $P' = P \times \{0\}$ and $Q' = \{0\} \times Q$. If $r_1 = 0$ then P is a single point and the statement is trivial, so assume $r_1, r_2 > 0$; then $P' \cap Q' = \emptyset$ and $|P' \cup Q'| = |P| + |Q|$. For $p \in P$ and $q \in Q$ we have $\|(p,0) - (0,q)\|^2 = \|p\|^2 + \|q\|^2 = r_1^2 + r_2^2 = 1$. Let T be a set of three points of $P' \cup Q'$. If T meets both P' and Q', then T contains a pair $\{(p,0),(0,q)\}$, which is at distance 1. Otherwise $T \subset P'$ or $T \subset Q'$, and T contains a pair at distance 1 because P and Q are almost-equidistant. □

In the normalised language, if $1/(1-c_1)+1/(1-c_2)=2$ then $f(d_1+d_2)\ge h(d_1,c_1)+h(d_2,c_2)$. Solving, $c_2=c_1/(2c_1-1)$; in particular exactly one of c_1,c_2 is positive unless $c_1=c_2=0$. The negative summand is controlled by the Bezdek–Lángi bound; the next lemma records what is attained by two cliques.

2.2. **Lemma** (Two cliques)**.** *Let $j\ge 1$ and $c\in(-1,0)$.*

(1) *If $c>-1/(j-1)$ (no condition if $j=1$), then $h(j,c)\ge 2j$.*
(2) *If $c=-1/j$, then $h(j,c)\ge 2j+2$.*

Proof. (1) The $j\times j$ matrix $G=(1-c)I+cJ$ has eigenvalues $1-c$ with multiplicity $j-1$ and $1+(j-1)c$ with multiplicity 1; both are positive when $c>-1/(j-1)$, so G is positive definite and is the Gram matrix of j unit vectors $u_1,\dots,u_j$ in $\mathbb{R}^j$ with $\langle u_a,u_b\rangle=c$ for $a\ne b$. Put $S=\{u_1,\dots,u_j\}$. The set $S\cup(-S)$ has $2j$ distinct elements: $u_a=-u_b$ would give $\langle u_a,u_b\rangle=-1<c$ if $a\ne b$ and $\|u_a\|=0$ if $a=b$. Any three elements of $S\cup(-S)$ contain two from S or two from $-S$, and those have inner product c. Hence $h(j,c)\ge 2j$.

(2) For $c=-1/j$ the $(j+1)\times(j+1)$ matrix $(1+1/j)I-(1/j)J$ is positive semidefinite of rank j (its eigenvalues are $1+1/j$, with multiplicity j, and 0), so it is the Gram matrix of $j+1$ unit vectors in $\mathbb{R}^j$ with pairwise inner products $-1/j$: the vertices of a regular simplex centred at the origin. As in (1), the union of this simplex and its antipodal image is a set of $2j+2$ unit vectors in $\mathbb{R}^j$ in which any three contain a pair with inner product $-1/j$. □

For $c=-1/3$, $j=3$, the set in (2) is the vertex set of a cube; this is the 8-point set used in [1] for $f(8)\ge 24$.

3. The spectral embedding

Proof of Theorem 1.1. Let A be the adjacency matrix of H, J the all-ones matrix and $\mathbf{1}$ the all-ones vector. Since H is k-regular and connected, $A\mathbf{1}=k\mathbf{1}$ and k is a simple eigenvalue, so $\lambda_2<k$. Let $u_1=\mathbf{1}/\sqrt{n},u_2,\dots,u_n$ be an orthonormal eigenbasis of A with eigenvalues $k=\lambda_1>\lambda_2\ge\lambda_3\ge\cdots\ge\lambda_n$. Define

$$M=\lambda_2 I-A+\frac{k-\lambda_2}{n}J.$$

Then $Mu_1=\lambda_2u_1-ku_1+(k-\lambda_2)u_1=0$, and for $i\ge 2$, $Ju_i=0$ (as $u_i\perp\mathbf{1}$) gives $Mu_i=(\lambda_2-\lambda_i)u_i$ with $\lambda_2-\lambda_i\ge 0$. Hence M is symmetric positive semidefinite, and its kernel is $\operatorname{span}(\mathbf{1})\oplus\ker(A-\lambda_2 I)$, of dimension $1+m$. Therefore $\operatorname{rank}M=n-1-m$, and there are vectors $p_1,\dots,p_n\in\mathbb{R}^{n-1-m}$ with $\langle p_i,p_j\rangle=\rho M_{ij}$ for all i,j, where $\rho=1/(2\lambda_2)>0$ (take the rows of $\sqrt{\rho}\,M^{1/2}$ restricted to the range of M).

The entries of M are $M_{ii}=\lambda_2+(k-\lambda_2)/n$, $M_{ij}=(k-\lambda_2)/n$ if $i\ne j$ are non-adjacent, and $M_{ij}=-1+(k-\lambda_2)/n$ if i,j are adjacent. Consequently

$$\|p_i-p_j\|^2=\rho(M_{ii}+M_{jj}-2M_{ij})=\begin{cases}2\rho\lambda_2=1, & i\not\sim j,\\ 2\rho(\lambda_2+1)=1+1/\lambda_2, & i\sim j,\end{cases}$$

and $\|p_i\|^2=\rho M_{ii}=\frac12+\frac{k-\lambda_2}{2n\lambda_2}=r^2$. The points are distinct since all distances are at least 1. If i,j,l are three vertices, they are not all pairwise adjacent because H is triangle-free, so some pair among p_i,p_j,p_l is at distance 1. Thus $\{p_1,\dots,p_n\}$ is almost-equidistant in $\mathbb{R}^{n-1-m}$, lies on the sphere of radius r, and realises only the distances 1 and $\sqrt{1+1/\lambda_2}$. □

Write $d=n-1-m$. The excess of the set over the general bound $2d+4$ is $n-2d-4=2m-n-2$, so Theorem 1.1 beats $2d+4$ exactly when λ_2 has multiplicity larger than $n/2+1$. For bipartite graphs the spectrum is symmetric and $m\le(n-2)/2$, so bipartite graphs never help; this is consistent with $h(d,0)=2d$.

3.1. **Corollary.** *Let H be a strongly regular graph with parameters $(v,k,0,\mu)$ (so H is triangle-free) and eigenvalues $k>\theta>\tau$ with multiplicities $1, m_\theta, m_\tau$. Then Theorem 1.1 applies with $n=v$, $\lambda_2=\theta$, $m=m_\theta$, and yields an almost-equidistant set of v points on a sphere of radius r in $\mathbb{R}^{m_\tau}$, where*

$$r^2=\frac{1}{2}+\frac{k-\theta}{2v\theta},\qquad c:=1-\frac{1}{2r^2}=\frac{k-\theta}{v\theta+k-\theta}.$$

The direct sum with the set of Lemma 2.2 then gives, with $q:=v\theta/(k-\theta)-1$,

$$f(m_\tau+j)\ge v+2j\quad(1\le j<q),\qquad f(m_\tau+q)\ge v+2q+2\quad\text{if } q\in\mathbb{Z}.$$

Proof. A strongly regular graph is connected with θ its second largest eigenvalue, $\theta>0$ since $\theta\tau=\mu-k<0$ and $\tau<0$; and $v-1-m_\theta=m_\tau$. The first statement is Theorem 1.1. For the second, the partner sphere has squared radius $1-r^2$, i.e. normalised inner product $c_2=c/(2c-1)=-(k-\theta)/(v\theta-k+\theta)=-1/q$. Lemma 2.2 gives $h(j,-1/q)\ge 2j$ for $j<q$ (as $-1/q>-1/(j-1)$) and $h(q,-1/q)\ge 2q+2$ when q is an integer, and Lemma 2.1 combines the two sets. □

Table 1 lists the data for the six known triangle-free strongly regular graphs other than the pentagon. (For the pentagon, $\theta=(\sqrt{5}-1)/2$ has multiplicity 2 and the theorem gives 5 points in $\mathbb{R}^2$, which is below the Moser spindle.) The Clebsch row is exactly the Larman–Rogers set and, with its partner cube, the 24-point set of [1]; these two rows serve as checks of the method against the literature.

graph	(v,k,λ,μ)	θ	τ	m_τ	r^2	c	q	bounds
Petersen	$(10,3,0,1)$	1	-2	4	3/5	1/6	4	$f(4)\ge 10$, $f(8)\ge 20$
Clebsch	$(16,5,0,2)$	1	-3	5	5/8	1/5	3	$f(5)\ge 16$, $f(8)\ge 24$
Hoffman–Singleton	$(50,7,0,1)$	2	-3	21	21/40	1/21	19	$f(21)\ge 50$, $f(40)\ge 90$
Gewirtz	$(56,10,0,2)$	2	-4	20	15/28	1/15	13	$f(20)\ge 56$, $f(33)\ge 84$
M_{22}	$(77,16,0,4)$	2	-6	21	6/11	1/12	10	$f(21)\ge 77$, $f(31)\ge 99$
Higman–Sims	$(100,22,0,6)$	2	-8	22	11/20	1/11	9	$f(22)\ge 100$, $f(31)\ge 120$

TABLE 1. The six triangle-free strongly regular graphs and the almost-equidistant sets of Corollary 3.1. The last column gives the standalone bound $f(m_\tau)\ge v$ and the direct sum with a double q-simplex.

For the Higman–Sims graph the intermediate values $1\le j\le 8$ of Corollary 3.1 give $f(22+j)\ge 100+2j$, which is the middle part of Theorem 1.2. The bound $f(10)\ge 26$ uses the other eigenspace of the Petersen graph. The orthogonal projection onto its θ-eigenspace ($\theta=1$, multiplicity 5) is $E_\theta=(4I+2A-J)/6$, with diagonal entries $1/2$ and off-diagonal entries $1/6$ (adjacent) and $-1/6$ (non-adjacent); it is positive semidefinite of rank 5, so $\frac{3}{4}E_\theta$ is the Gram matrix of 10 points $p_1,\dots,p_{10}\in\mathbb{R}^5$ with $\|p_i\|^2=3/8$ and $\|p_i-p_j\|^2=\frac{3}{4}(1-2(E_\theta)_{ij})$, which equals 1 for non-adjacent and $1/2$ for adjacent pairs. The non-adjacent pairs of the Petersen graph form the triangular graph $T(5)$, whose complement is triangle-free, so this 10-point set is almost-equidistant. Since $3/8+5/8=1$, Lemma 2.1 combines it with the 16-point Clebsch set (squared radius $5/8$) in an orthogonal $\mathbb{R}^5$, giving 26 points in $\mathbb{R}^{10}$.

4. Extension points from Delsarte-tight cocliques

Let H be a strongly regular graph with parameters $(v,k,0,\mu)$ and eigenvalues $k>\theta>\tau$, and let $P=\{p_1,\dots,p_v\}\subset\mathbb{R}^{m_\tau}$ be the set of Corollary 3.1. For a coclique C of H, the Delsarte–Hoffman bound states $|C|\le v(-\tau)/(k-\tau)$, with equality if and only if every vertex outside C has exactly $-\tau$ neighbours in C (see [7, Section 9.6] or [5]). We call such cocliques *tight*.

4.1. **Lemma.** *With the notation above, let $D \ge m_\tau$, regard P as a subset of $\mathbb{R}^{m_\tau} \subseteq \mathbb{R}^D$, let $x \in \mathbb{R}^D \setminus P$ and $C = \{i : \|x - p_i\| \ne 1\}$.*

(1) *If $P \cup \{x\}$ is almost-equidistant, then either $C = \emptyset$ (and then x is orthogonal to $\operatorname{span} P$ with $\|x\|^2 = 1 - r^2$), or C is a tight coclique of H.*
(2) *Conversely, for every tight coclique C there are exactly two points $x^+, x^- \in \mathbb{R}^{m_\tau}$ with $\{i : \|x^\pm - p_i\| \ne 1\} \subseteq C$, and both $P \cup \{x^+\}$ and $P \cup \{x^-\}$ are almost-equidistant sets of $v+1$ points in $\mathbb{R}^{m_\tau}$.*

Proof. We use the realisation from the proof of Theorem 1.1: $p_i = \sqrt{\rho}\, M^{1/2} e_i$ with $\rho = 1/(2\theta)$ and $M = \theta I - A + \frac{k-\theta}{v} J$. The range W of M is the τ-eigenspace of A (the θ-eigenspace and $\mathbf{1}$ span the kernel of M, and M acts on the τ-eigenspace as multiplication by $\theta - \tau$); P spans W, and we identify $\mathbb{R}^{m_\tau}$ with W. Let $U = \{1, \dots, v\} \setminus C$.

(1) If $i, j \in C$ were adjacent, the triple $\{x, p_i, p_j\}$ would contain no pair at distance 1 (adjacent pairs are at distance $\sqrt{1 + 1/\theta}$); so C is a coclique. Write $x = x' + x''$ with $x' \in W$ and $x'' \perp W$. For $i \in U$, the condition $\|x - p_i\|^2 = 1$ reads $\|x\|^2 - 2\langle x', p_i\rangle + r^2 = 1$, so

$$\langle x', p_i\rangle = \alpha := \tfrac{1}{2}(\|x\|^2 + r^2 - 1) \qquad \text{for all } i \in U.$$

Put $z = \sqrt{\rho}\, M^{1/2} x' \in W$; then $\langle x', p_i\rangle = \langle x', \sqrt{\rho}\, M^{1/2} e_i\rangle = z_i$, so z is a τ-eigenvector of A that equals α on U.

Case $\alpha = 0$. Then z vanishes on U. For $i \in C$ all neighbours of i lie in U (as C is a coclique), hence $\tau z_i = (Az)_i = \sum_{j \sim i} z_j = 0$ and $z_i = 0$. So $z = 0$, hence $x' = 0$ (as $M^{1/2}$ is injective on W), hence $\langle x, p_i\rangle = 0$ for all i and $\|x - p_i\|^2 = \|x\|^2 + r^2$ is the same for all i. Therefore $C = \emptyset$ or $C = \{1, \dots, v\}$; the latter is not a coclique since H has edges. Thus $C = \emptyset$, $x = x'' \perp W$ and $\|x\|^2 = 1 - r^2$.

Case $\alpha \ne 0$. Let $y = z/\alpha$, a τ-eigenvector equal to 1 on U. For $i \in C$, all neighbours of i are in U, so $\tau y_i = (Ay)_i = k$ and $y_i = k/\tau$. For $i \in U$ let $t_i = |N(i) \cap C|$; then $\tau = (Ay)_i = (k - t_i) + t_i\, k/\tau$, i.e. $t_i(k/\tau - 1) = \tau - k$, i.e. $t_i = -\tau$. In particular $C \ne \emptyset$ (as $\tau \ne 0$). Counting the edges between C and U gives $k|C| = -\tau\,|U| = -\tau(v - |C|)$, so $|C| = v(-\tau)/(k - \tau)$ and C is tight.

(2) Let C be a tight coclique and $y = \mathbf{1}_U + \frac{k}{\tau}\mathbf{1}_C$. For $i \in C$, $(Ay)_i = k = \tau y_i$; for $i \in U$, $(Ay)_i = (k + \tau)\cdot 1 + (-\tau)\cdot k/\tau = \tau = \tau y_i$. So $y \in W$ and $M^{1/2} y = \sqrt{\theta - \tau}\, y$. For real γ put

$$x_\gamma = \frac{\gamma}{\sqrt{\rho(\theta - \tau)}}\, y \in W, \qquad \text{so that} \qquad \langle x_\gamma, p_i\rangle = \sqrt{\rho}\,(M^{1/2} x_\gamma)_i = \gamma\, y_i.$$

For $i \in U$ the condition $\|x_\gamma - p_i\|^2 = 1$ becomes $\|x_\gamma\|^2 - 2\gamma + r^2 = 1$, i.e.

$$\frac{\|y\|^2}{\rho(\theta - \tau)}\, \gamma^2 - 2\gamma + (r^2 - 1) = 0. \tag{2}$$

Since $r^2 < 1$, the product of the roots of (2) is negative, so it has two real roots $\gamma^+ > 0 > \gamma^-$; put $x^\pm = x_{\gamma^\pm}$. Then $\|x^\pm - p_i\| = 1$ for all $i \in U$. Any three points of $P \cup \{x^\pm\}$ contain a pair at distance 1: if none of them is $x^\pm$ this holds because P is almost-equidistant; if the triple is $\{x^\pm, p_i, p_j\}$ with $i \in U$ or $j \in U$, the corresponding pair is at distance 1; and if $i, j \in C$ then $i \not\sim j$ and $\|p_i - p_j\| = 1$. Next, $x^\pm \notin P$: for $j \in U$ the inner products $\langle x^\pm, p_j\rangle = \gamma^\pm$ are constant, whereas for every i the inner products $\langle p_i, p_j\rangle$, $j \in U \setminus \{i\}$, take both values $\rho(k - \theta)/v$ (non-adjacent) and $\rho(k - \theta)/v - \rho$ (adjacent): a vertex $i \in C$ has its $k \ge 1$ neighbours in U and $|U| - k \ge 1$ non-neighbours in U, and a vertex $i \in U$ has $k + \tau \ge 1$ neighbours and $|U| - 1 - (k + \tau) \ge 1$ non-neighbours in $U \setminus \{i\}$ (these inequalities hold for all graphs in Table 1, and are checked by the script). Finally, if $x \in W$ satisfies $\{i : \|x - p_i\| \ne 1\} \subseteq C$, then part (1) applied with $x'' = 0$ shows $x = x_\alpha$ with α a root of (2) (indeed $z = \alpha y$ because $z - \alpha y$ is a τ-eigenvector vanishing on U, hence zero by the argument of the case $\alpha = 0$), so $x \in \{x^+, x^-\}$. □

4.2. **Remark.** The proof of the first part is written with the degenerate case $x \perp W$ spelled out because it is exactly the mechanism behind the bounds $f(6) \geq 18$ and $f(7) \geq 20$ of [1]: a point orthogonal to W at distance $\sqrt{1-r^2}$ from the origin is at distance 1 from all of P, and Lemma 2.1 is the systematic version of this.

Two extension points x_1, x_2 coming from tight cocliques C_1, C_2 can be added simultaneously provided every triple $\{x_1, x_2, p_i\}$ contains a unit pair, i.e. provided $C_1 \cap C_2 = \emptyset$ or $\|x_1 - x_2\| = 1$ (and $x_1 \neq x_2$); three or more extension points additionally need a unit pair among themselves. Which of these happen is a finite exact computation:

4.3. **Proposition.** (1) *The Petersen graph has* 5 *tight cocliques (the* 4*-sets of pairs through a fixed element of* $\{1, \dots, 5\}$*). For two of them,* $C_1 \neq C_2$*, the points* x_1^+ *and* x_2^- *are at distance* 1*; hence* $f(4) \geq 12$*.*

(2) *The Hoffman–Singleton graph has* 100 *tight cocliques of size* 15*, among which there are disjoint pairs; hence* $f(21) \geq 52$*.*

(3) *The Gewirtz graph has* 42 *tight cocliques of size* 16*, among which there are disjoint pairs; hence* $f(20) \geq 58$*.*

(4) *The* M_{22} *graph has* 22 *tight cocliques of size* 21 *(the blocks of* $S(3, 6, 22)$ *through a point), any two of which meet; hence* $f(21) \geq 78$*.*

(5) *The Clebsch and Higman–Sims graphs have no tight cocliques (*$v(-\tau)/(k-\tau) = 6 > \alpha = 5$ *and* $80/3 \notin \mathbb{Z}$ *respectively), so their sets admit no extension point in any dimension.*

In cases (2)–(4), an exhaustive check of all pairs and signs shows that no larger family of extension points can be added to P *inside* $\mathbb{R}^{m_\tau}$*.*

Proof. The enumeration of tight cocliques and the distance computations are performed exactly by the script in Appendix A (the distances $\|x_1 - x_2\|^2$ lie in the real quadratic field generated by the discriminant of (2), and the script computes in that field). The almost-equidistant property of each resulting set is checked triple by triple. Item (1) recovers the known value $f(4) \geq 12$ [1]. The last sentence is the outcome of the exhaustive search over all pairs of tight cocliques and both roots, which we performed with floating-point arithmetic at tolerance 10^{-9}; since by Lemma 4.1 every extension point in $\mathbb{R}^{m_\tau}$ is one of the $x_C^\pm$, this search is exhaustive. □

Extension points lie off the sphere of P and therefore cannot be combined with Lemma 2.1; this is why $f(23) \geq 102$ uses the Higman–Sims set with two antipodal points rather than the 78-point set. Iterating the extension idea one step further (adjoining to P all x_C^+ for the 22 point-cocliques of M_{22}, after allowing one extra coordinate so that the new points can be made pairwise at distance 1) reproduces the Higman–Sims set: the tower Gewirtz $\subset M_{22} \subset$ Higman–Sims is the tower of extensions, and the Higman–Sims set is its top.

5. Proof of Theorem 1.2 and remarks

Proof of Theorem 1.2. $f(10) \geq 26$ was established after Table 1; $f(20) \geq 58$, $f(21) \geq 78$ are Proposition 4.3(3),(4); $f(22) \geq 100$ is the Higman–Sims row of Table 1; $f(22+j) \geq 100 + 2j$ for $1 \leq j \leq 8$ and $f(31) \geq 120$ are Corollary 3.1 for the Higman–Sims graph ($q = 9$). Monotonicity $f(d+1) \geq f(d)$ is clear (embed $\mathbb{R}^d \subset \mathbb{R}^{d+1}$), and $120 > 2d + 4$ for $d \leq 57$. Each of these point sets (and the checks $f(4) \geq 12$, $f(5) \geq 16$, $f(8) \geq 24$ against the literature, twenty-five sets in all) is verified exactly by the script in Appendix A, whose printed summary includes

d	4	5	8	10	20	21	22	23	24	25	26	27	28	29	30	31
n	12	16	24	26	58	78	100	102	104	106	108	110	112	114	116	120

□

5.1. **Remark** (On the ratio $f(d)/d$)**.** The best ratios $f(d)/d$ now known are $100/22 \approx 4.55$ at $d = 22$ and $120/31 \approx 3.87$ at $d = 31$, compared with $16/5 = 3.2$ at $d = 5$ previously. Polyanskii [13] showed that almost-equidistant sets contained in a ball of radius $1/\sqrt{2} + cd^{-2/3}$, $c < 1/2$, have $O(d)$ points; the sets of Theorem 1.1 live on spheres of radius $r > 1/\sqrt{2}$ with $r^2 - \frac{1}{2} = \frac{k-\lambda_2}{2n\lambda_2}$, e.g. $r^2 = 11/20$ for Higman–Sims, well outside that regime.

5.2. **Remark** (Asymptotics)**.** Theorem 1.1 together with Corollary 3.1 shows that a triangle-free strongly regular graph with $2m_\theta > v$ gives $f(d) > 2d + 4$ in the two dimensions $d = m_\tau$ and $d = m_\tau + q$. All seven known triangle-free strongly regular graphs have $2m_\theta > v$. Whether infinitely many triangle-free strongly regular graphs exist is a well-known open problem [5]; if they do, and if $2m_\theta > v$ persists, then $f(d) > 2d+4$ for infinitely many d. More generally, the question whether $f(d)/d$ is bounded is equivalent, for two-distance sets, to Polyanskii's Problem 19: it asks for triangle-free graphs H on n vertices with $\operatorname{rank}(A_H - \lambda_2 I) = o(n)$ and $\lambda_2 > 0$; the Higman–Sims graph gives rank $= 23$ at $n = 100$.

5.3. **Remark** (Scope of the searches)**.** The following statements are the outcome of computer searches, and we do not claim optimality beyond them. (i) A search over all triangle-free Cayley graphs on cyclic groups of order at most 34 and over random maximal sum-free connection sets in $\mathbb{Z}_2^k$, $k \le 7$, found no triangle-free Cayley graph on n vertices with $\lambda_2 > 0$ of multiplicity larger than $n/2 + 1$ other than the Clebsch graph (the folded 5-cube); bipartite and $2d$-type examples appear in abundance. (ii) For the partner spheres relevant to Table 1 we enumerated, for $(d, c) \in \{(4, -\frac{1}{3}), (6, -\frac{1}{3}), (7, -\frac{1}{3}), (3, -\frac{1}{9}), (4, -\frac{1}{9})\}$, all triangle-free graphs on $2d+1$ and $2d+2$ vertices (for $c = -\frac{1}{3}$ and $d \ge 6$, on 12 and 13 vertices, the most that $R(3, 5) = 14$ allows) whose complement has clique number at most the maximum size of a c-clique, and attempted to realise the complement of each edge-maximal one as unit vectors in $\mathbb{R}^d$ with inner product c on its edges by quasi-Newton minimisation from 20–40 random starts; every attempt stalled at residual at least $3 \cdot 10^{-3}$. Within this search we therefore found no 9-point set for $(4, -\frac{1}{3})$, no 7-point set for $(3, -\frac{1}{9})$, no 9-point set for $(4, -\frac{1}{9})$, and no 11-point set for $(6, -\frac{1}{3})$, i.e. nothing beyond the two-clique sets of Lemma 2.2; we do not claim that these values of h are exact. This is why Theorem 1.2 has $100 + 2j$ rather than $100 + 2j + 1$ for $23 \le d \le 30$, and why $f(9)$ and $f(11)$ are not improved. (iii) A simulated-annealing search for large almost-equidistant subsets of the root systems D_4–D_8, E_6, E_7, E_8, of $\{\pm 1\}^k$ ($k \le 7$) and of $\{0, \pm 1\}^5$, for every squared distance occurring, found nothing exceeding the previously known bounds in dimensions 4–9 (the largest were 17 points of affine dimension 6 in E_7 and 21 points of affine dimension 8 in E_8, both below 18 and 24). The code for all searches accompanies the paper.

5.4. **Question.** Is $f(d)$ bounded by Cd for an absolute constant C? The constructions here give $C \ge 100/22$. Is there a 101-point almost-equidistant set in $\mathbb{R}^{22}$? Lemma 4.1 shows that none is obtained by adding a point to the Higman–Sims set.

Appendix A. Exact verification script

The script below (Python 3, standard library only; also provided as an ancillary file[1]) constructs the six graphs from the extended binary Golay code and the Hoffman–Singleton pentagon–pentagram model, verifies the strongly regular identities $A^2 = kI + \mu(J - I - A)$, verifies that $N = (A - \theta I)(A - kI)$ satisfies $N^2 = (\tau - \theta)(\tau - k)N$ (so that $N/((\tau - \theta)(\tau - k))$ is the orthogonal projection onto the τ-eigenspace, hence positive semidefinite of rank m_τ; this is $M/(k - \tau)$ in the notation of Theorem 1.1), forms the exact rational Gram matrices of all point sets, computes

[1]Code and data accompany the paper as arXiv ancillary files: `verify_certificate.py`, `reproduce.py` (numpy; writes explicit coordinates to `certificate.json`), and the search codes `cayley.py`, `gen.py`, `sphsearch.py`, `lattice.py`, `hoffman.py`.

the extension points in the quadratic field of (2), and checks every triple. It runs in about one minute.

```
#!/usr/bin/env python3
"""Standalone, dependency-free (Python 3 standard library only) verification of the
almost-equidistant sets constructed in the paper.

Every step is exact: integer arithmetic for graphs and idempotents, Fractions for
Gram matrices, and arithmetic in a real quadratic field for the extension points.

A finite set is almost-equidistant if among any three of its points some two are
at distance 1.  We certify point sets by their Gram matrices G (entries <p_i,p_j>):
  * realisability in R^d  <=>  G symmetric PSD of rank <= d  (Cholesky/spectral thm),
    which we prove structurally (G is a positive multiple of a symmetric idempotent,
    or a block-diagonal / bordered version thereof);
  * |p_i - p_j|^2 = G_ii + G_jj - 2 G_ij, checked exactly;
  * the almost-equidistant property is checked over all triples.
"""
import itertools, sys
from fractions import Fraction as Fr

# ---------------------------------------------------------------------- Golay code
def golay_octads():
    n = 23
    g = [1, 0, 1, 0, 1, 1, 1, 0, 0, 0, 1, 1]       # x^11+x^10+x^6+x^5+x^4+x^2+1
    words = set()
    for m in range(1 << 12):
        cw = [0] * n
        for i in range(12):
            if (m >> i) & 1:
                for j, gj in enumerate(g):
                    if gj: cw[(i + j) % n] ^= 1
        words.add(tuple(cw + [sum(cw) & 1]))
    assert len(words) == 4096
    wd = {}
    for w in words: wd[sum(w)] = wd.get(sum(w), 0) + 1
    assert wd == {0: 1, 8: 759, 12: 2576, 16: 759, 24: 1}
    return [frozenset(i for i in range(24) if w[i]) for w in words if sum(w) == 8]

def steiner_3_6_22():
    blocks = [frozenset(o - {22, 23}) for o in golay_octads() if 22 in o and 23 in o]
    assert len(blocks) == 77
    cnt = {}
    for b in blocks:
        for t in itertools.combinations(sorted(b), 3): cnt[t] = cnt.get(t, 0) + 1
    assert len(cnt) == 1540 and set(cnt.values()) == {1}
    return blocks

# ---------------------------------------------------------------------- graphs
def zeros(n): return [[0] * n for _ in range(n)]

def higman_sims(B):
    A = zeros(100)
    for p in range(22): A[0][1 + p] = A[1 + p][0] = 1
    for i, b in enumerate(B):
        for p in b: A[1 + p][23 + i] = A[23 + i][1 + p] = 1
        for j in range(i + 1, 77):
            if not (b & B[j]): A[23 + i][23 + j] = A[23 + j][23 + i] = 1
    return A

def disjointness_graph(B):
    n = len(B); A = zeros(n)
    for i in range(n):
        for j in range(i + 1, n):
```

```
            if not (B[i] & B[j]): A[i][j] = A[j][i] = 1
    return A

def hoffman_singleton():
    A = zeros(50)
    P = lambda h, i: 5 * h + (i % 5); Q = lambda j, i: 25 + 5 * j + (i % 5)
    for h in range(5):
        for i in range(5):
            A[P(h, i)][P(h, i + 1)] = A[P(h, i + 1)][P(h, i)] = 1
            A[Q(h, i)][Q(h, i + 2)] = A[Q(h, i + 2)][Q(h, i)] = 1
            for j in range(5):
                A[P(h, i)][Q(j, h * j + i)] = A[Q(j, h * j + i)][P(h, i)] = 1
    return A

def clebsch():
    A = zeros(16)
    for x in range(16):
        for s in (1, 2, 4, 8, 15): A[x][x ^ s] = 1
    return A

def petersen():
    V = list(itertools.combinations(range(5), 2)); A = zeros(10)
    for i in range(10):
        for j in range(10):
            if i != j and not (set(V[i]) & set(V[j])): A[i][j] = 1
    return A

# ---------------------------------------------------------------- matrices
def matmul(X, Y):
    n = len(X); m = len(Y[0]); k = len(Y)
    return [[sum(X[i][t] * Y[t][j] for t in range(k)) for j in range(m)] for i in range(n)]

def verify_srg(A, v, k, lam, mu):
    n = len(A); assert n == v
    for i in range(n):
        assert A[i][i] == 0 and sum(A[i]) == k
        for j in range(n): assert A[i][j] == A[j][i] in (0, 1)
    A2 = matmul(A, A)
    for i in range(n):
        for j in range(n):
            exp = k if i == j else (lam if A[i][j] else mu)
            assert A2[i][j] == exp
    # triangle-free  <=>  lam == 0
    assert lam == 0

def srg_spectrum(v, k, lam, mu):
    disc = (lam - mu) ** 2 + 4 * (k - mu); s = int(round(disc ** 0.5)); assert s * s == disc
    theta = ((lam - mu) + s) // 2; tau = ((lam - mu) - s) // 2
    m_theta = ((v - 1) * (-tau) - k) // (theta - tau); m_tau = v - 1 - m_theta
    assert k + m_theta * theta + m_tau * tau == 0
    return theta, tau, m_theta, m_tau

def tau_projection(A, v, k, lam, mu):
    """Return integer N = (A - theta I)(A - k I) and s with N^2 = s N, so E = N/s is a
    symmetric idempotent (orthogonal projection onto the tau-eigenspace), rank m_tau."""
    theta, tau, m_theta, m_tau = srg_spectrum(v, k, lam, mu)
    n = len(A)
    M1 = [[A[i][j] - (theta if i == j else 0) for j in range(n)] for i in range(n)]
    M2 = [[A[i][j] - (k if i == j else 0) for j in range(n)] for i in range(n)]
    N = matmul(M1, M2); s = (tau - theta) * (tau - k)
    N2 = matmul(N, N)
    for i in range(n):
        for j in range(n): assert N2[i][j] == s * N[i][j] and N[i][j] == N[j][i]
    assert sum(N[i][i] for i in range(n)) == s * m_tau
```

```
    return N, s, m_tau, theta, tau

# ------------------------------------------------------------------ point sets
class PointSet:
    """A finite point set given by an exact Gram matrix (list of lists of numbers),
    a proof-of-realisability dimension d, and a description."""
    def __init__(self, G, d, name):
        self.G, self.d, self.name = G, d, name
    def dist2(self, i, j):
        return self.G[i][i] + self.G[j][j] - 2 * self.G[i][j]
    def check_almost_equidistant(self):
        n = len(self.G)
        unit = [[self.dist2(i, j) == 1 for j in range(n)] for i in range(n)]
        for i in range(n):
            for j in range(i + 1, n):
                assert self.dist2(i, j) != 0, "coincident points"
        for a, b, c in itertools.combinations(range(n), 3):
            assert unit[a][b] or unit[a][c] or unit[b][c], "bad triple %s" % ((a, b, c),)
        return n

def srg_point_set(name, A, par):
    """Points P_i = rho E e_i on the sphere |P|^2 = r^2 in the tau-eigenspace (dim m_tau),
    scaled so that non-adjacent pairs are at distance 1.  Gram = rho^2 E."""
    verify_srg(A, *par)
    N, s, m_tau, theta, tau = tau_projection(A, *par)
    v = par[0]
    E = [[Fr(N[i][j], s) for j in range(v)] for i in range(v)]
    # non-adjacent pairs have E_ij = e_na; want 2 rho^2 (E_ii - e_na) = 1
    i0, j0 = next((i, j) for i in range(v) for j in range(v) if i != j and A[i][j] == 0)
    rho2 = 1 / (2 * (E[0][0] - E[i0][j0]))
    G = [[rho2 * E[i][j] for j in range(v)] for i in range(v)]
    P = PointSet(G, m_tau, name)
    P.E, P.rho2, P.r2, P.A, P.par, P.tau, P.k = E, rho2, rho2 * E[0][0], A, par, tau, par[1]
    return P

def direct_sum(P, Q):
    """P on sphere |p|^2 = r1^2 in R^d1, Q on sphere |q|^2 = r2^2 in R^d2, r1^2+r2^2 = 1."""
    assert P.r2 + Q.r2 == 1
    n1, n2 = len(P.G), len(Q.G)
    G = [[Fr(0)] * (n1 + n2) for _ in range(n1 + n2)]
    for i in range(n1):
        for j in range(n1): G[i][j] = P.G[i][j]
    for i in range(n2):
        for j in range(n2): G[n1 + i][n1 + j] = Q.G[i][j]
    R = PointSet(G, P.d + Q.d, P.name + " (+) " + Q.name)
    return R

def double_simplex(d):
    """Vertices of a regular unit simplex centred at the origin in R^d together with
    their antipodes: 2d+2 points on the sphere |q|^2 = d/(2(d+1))."""
    r2 = Fr(d, 2 * (d + 1)); n = d + 1
    # Gram of S: r2 * ((1+1/d) I - (1/d) J);  antipodes: -S
    G = [[Fr(0)] * (2 * n) for _ in range(2 * n)]
    for i in range(2 * n):
        for j in range(2 * n):
            a, b = i % n, j % n
            g = r2 * ((1 + Fr(1, d)) if a == b else 0) - r2 * Fr(1, d)
            if (i < n) != (j < n): g = -g
            G[i][j] = g
    Q = PointSet(G, d, "double %d-simplex" % d); Q.r2 = r2
    return Q

def two_cliques(d, c2):
    """Two regular d-point sets of unit vectors with pairwise inner product c2 (<0, 1/|c2|>d),
```

```
    scaled to the sphere |q|^2 = r2 = 1/(2(1-c2)); rank d each; placed in the same R^d."""
    r2 = 1 / (2 * (1 - c2))
    # first clique: Gram r2*((1-c2) I + c2 J) on d points (rank d since 1 - c2 + d c2 > 0)
    assert 1 - c2 + d * c2 > 0
    # second clique: the antipodal copy -S (cross inner products -c2 or -1)
    n = d; G = [[Fr(0)] * (2 * n) for _ in range(2 * n)]
    for i in range(2 * n):
        for j in range(2 * n):
            a, b = i % n, j % n
            g = r2 * (1 if a == b else c2)
            if (i < n) != (j < n): g = -g
            G[i][j] = g
    Q = PointSet(G, d, "two %d-cliques (c=%s)" % (d, c2)); Q.r2 = r2
    # cross distances must not be 1: 2 r2 (1 + c2) != 1 and 4 r2 != 1  (checked by triple test anyway)
    return Q

# ---------------------------------------------- quadratic field Q(sqrt(D))
class Qs:
    """a + b*sqrt(D) with a,b in Q, D a fixed positive non-square rational."""
    D = None
    def __init__(self, a, b=0): self.a, self.b = Fr(a), Fr(b)
    def __add__(s, o): o = o if isinstance(o, Qs) else Qs(o); return Qs(s.a + o.a, s.b + o.b)
    __radd__ = __add__
    def __neg__(s): return Qs(-s.a, -s.b)
    def __sub__(s, o): return s + (-(o if isinstance(o, Qs) else Qs(o)))
    def __rsub__(s, o): return Qs(o) - s
    def __mul__(s, o):
        o = o if isinstance(o, Qs) else Qs(o)
        return Qs(s.a * o.a + s.b * o.b * Qs.D, s.a * o.b + s.b * o.a)
    __rmul__ = __mul__
    def __eq__(s, o): o = o if isinstance(o, Qs) else Qs(o); return s.a == o.a and s.b == o.b
    def __ne__(s, o): return not s.__eq__(o)
    def __hash__(s): return hash((s.a, s.b))
    def sign(s):        # sign of a + b sqrt(D)
        if s.b == 0: return (s.a > 0) - (s.a < 0)
        if s.a == 0: return (s.b > 0) - (s.b < 0)
        t = s.a * s.a - s.b * s.b * Qs.D   # sign(a+b sqrt D) = sign(a) if a^2 > b^2 D else sign(b)
        return ((s.a > 0) - (s.a < 0)) if t > 0 else ((s.b > 0) - (s.b < 0))
    def __repr__(s): return "(%s + %s*sqrt(%s))" % (s.a, s.b, Qs.D)

def hoffman_cocliques(A, v, k, tau):
    """All cocliques C with |C| = v(-tau)/(k-tau) such that every vertex outside C has
    exactly -tau neighbours in C (Delsarte/Hoffman-tight cocliques)."""
    s_num, s_den = v * (-tau), k - tau
    if s_num % s_den: return []
    s = s_num // s_den
    nb = [sum(1 << j for j in range(v) if A[i][j]) for i in range(v)]
    out = []
    def rec(chosen, cand, cnt):
        if cnt == s: out.append(chosen); return
        if cnt + bin(cand).count("1") < s: return
        c = cand
        while c:
            i = (c & -c).bit_length() - 1; c &= c - 1
            rec(chosen | (1 << i), c & ~nb[i], cnt + 1)
    rec(0, (1 << v) - 1, 0)
    hoff = []
    for m in out:
        C = [i for i in range(v) if m >> i & 1]
        if all(sum(A[i][j] for j in C) == -tau for i in range(v) if not (m >> i & 1)): hoff.append(C)
    return hoff

def extend(P, cocliques, signs=None):
    """Add one extension point per Hoffman coclique C (with the '+' root).
```

```
    x_C = beta * y_C with y_C = 1_U + (k/tau) 1_C, a tau-eigenvector; x_C lies in the span
    of the P_i, so the dimension does not increase.  Gram entries:
      <x_C, P_i> = beta*rho*y_i,  <x_C, x_C'> = beta^2 <y_C, y_C'>,
    beta determined by |x_C - P_i|^2 = 1 for i in U.  Work in Q(sqrt(disc))."""
    v = len(P.G); k, tau = P.k, P.tau
    rho2 = P.rho2; r2 = P.r2
    ys = []
    for C in cocliques:
        y = [Fr(k, tau) if i in C else Fr(1) for i in range(v)]
        # check y is a tau-eigenvector of A (so E y = y):
        for i in range(v):
            assert sum(P.A[i][j] * y[j] for j in range(v)) == tau * y[i]
        ys.append(y)
    y2 = sum(t * t for t in ys[0])
    # beta^2 y2 - 2 beta rho + r2 = 1.  Put gamma = beta*rho:  (y2/rho2) gamma^2 - 2 gamma + (r2-1) = 0
    # gamma = (1 + sqrt(1 - (y2/rho2)(r2-1))) / (y2/rho2)
    q = y2 / rho2
    Qs.D = 1 - q * (r2 - 1); assert Qs.D > 0
    if signs is None: signs = [1] * len(cocliques)
    gam = {sg: Qs(Fr(1, 1) / q, Fr(sg, 1) / q) for sg in (1, -1)}   # (1 +- sqrt(D))/q
    n = v + len(cocliques)
    G = [[Qs(0)] * n for _ in range(n)]
    for i in range(v):
        for j in range(v): G[i][j] = Qs(P.G[i][j])
    for a, y in enumerate(ys):
        ga = gam[signs[a]]
        for i in range(v): G[v + a][i] = G[i][v + a] = ga * y[i]
        for b, y2v in enumerate(ys):
            gb = gam[signs[b]]
            G[v + a][v + b] = ga * gb * (1 / rho2) * sum(y[i] * y2v[i] for i in range(v))
    R = PointSet(G, P.d, P.name + " + %d extension point(s)" % len(cocliques))
    R.qD = Qs.D
    return R

# ---------------------------------------------------------------------- main
def main():
    B = steiner_3_6_22()
    specs = [("Petersen", petersen(), (10, 3, 0, 1)),
             ("Clebsch", clebsch(), (16, 5, 0, 2)),
             ("Hoffman-Singleton", hoffman_singleton(), (50, 7, 0, 1)),
             ("Gewirtz", disjointness_graph([b for b in B if 0 not in b]), (56, 10, 0, 2)),
             ("M22", disjointness_graph(B), (77, 16, 0, 4)),
             ("Higman-Sims", higman_sims(B), (100, 22, 0, 6))]
    sets = {}
    results = []
    def report(P):
        n = P.check_almost_equidistant()
        results.append((P.d, n, P.name)); print("  f(%d) >= %d : %s" % (P.d, n, P.name)); return n
    print("Strongly regular graphs and their tau-eigenspace embeddings:")
    for name, A, par in specs:
        P = srg_point_set(name, A, par); sets[name] = P
        j0 = next(j for j in range(1, len(A)) if A[0][j] == 0)
        print("  %-18s SRG%s tau=%d m_tau=%d r^2=%s c_unit=%s"
              % (name, par, P.tau, P.d, P.r2, P.E[0][j0] / P.E[0][0]))
        report(P)
    print("Direct sums with a double simplex (partner inner product -1/d2):")
    for name, d2 in [("Clebsch", 3), ("Petersen", 4), ("Hoffman-Singleton", 19),
                     ("Gewirtz", 13), ("M22", 10), ("Higman-Sims", 9)]:
        report(direct_sum(sets[name], double_simplex(d2)))
    print("Higman-Sims (+) two j-cliques at inner product -1/9, j = 1..8:")
    for j in range(1, 9):
        report(direct_sum(sets["Higman-Sims"], two_cliques(j, Fr(-1, 9))))
    print("Petersen in its theta-eigenspace (unit pairs at inner product -1/3) (+) Clebsch:")
    # Petersen theta-embedding: E_theta = (A - tau I)(A - k I)/((theta-tau)(theta-k)); use complement trick:
```

```
    A = petersen(); v, k, lam, mu = 10, 3, 0, 1
    theta, tau, mth, mt = srg_spectrum(v, k, lam, mu)
    M1 = [[A[i][j] - (tau if i == j else 0) for j in range(v)] for i in range(v)]
    M2 = [[A[i][j] - (k if i == j else 0) for j in range(v)] for i in range(v)]
    N = matmul(M1, M2); s = (theta - tau) * (theta - k); N2 = matmul(N, N)
    for i in range(v):
        for j in range(v): assert N2[i][j] == s * N[i][j]
    assert sum(N[i][i] for i in range(v)) == s * mth
    E = [[Fr(N[i][j], s) for j in range(v)] for i in range(v)]
    i0, j0 = next((i, j) for i in range(v) for j in range(v) if i != j and A[i][j] == 0)
    rho2 = 1 / (2 * (E[0][0] - E[i0][j0]))
    Pt = PointSet([[rho2 * E[i][j] for j in range(v)] for i in range(v)], mth,
                  "Petersen (theta-space, r^2=%s)" % (rho2 * E[0][0]))
    Pt.r2 = rho2 * E[0][0]; assert Pt.r2 == Fr(3, 8)
    report(direct_sum(sets["Clebsch"], Pt))
    print("Extension points from Hoffman cocliques:")
    for name, want in [("Petersen", 2), ("Hoffman-Singleton", 2), ("Gewirtz", 2), ("M22", 1)]:
        P = sets[name]; v = len(P.G)
        H = hoffman_cocliques(P.A, v, P.k, P.tau)
        # choose pairwise disjoint cocliques greedily up to 'want'
        chosen = []
        for C in H:
            if all(not (set(C) & set(D)) for D in chosen): chosen.append(C)
            if len(chosen) == want: break
        signs = [1] * len(chosen)
        if len(chosen) < want:      # Petersen: cocliques meet; use the two roots of opposite sign
            chosen = H[:want]; signs = [1, -1]
        print("  %s: %d Hoffman cocliques of size %d; using %d" % (name, len(H), len(H[0]), len(chosen)))
        report(extend(P, chosen, signs))
    for name in ("Clebsch", "Higman-Sims"):
        P = sets[name]; assert hoffman_cocliques(P.A, len(P.G), P.k, P.tau) == [], name
    print("  (Clebsch and Higman-Sims have no Hoffman cocliques: no extension point exists.)")
    best = {}
    for d, n, nm in results:
        if n > best.get(d, (0,))[0]: best[d] = (n, nm)
    print("\nSummary of certified lower bounds f(d) >= n  (published: 2d+4 for d>=10, 24 for d=8,9):")
    for d in sorted(best):
        n, nm = best[d]
        print("  d=%2d  n=%3d  (2d+4=%3d)  %s" % (d, n, 2 * d + 4, nm))
    print("ALL CHECKS PASSED")

if __name__ == "__main__":
    main()
```

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